

STANDARD - NORMAL DISTRIBUTION :-

↳ mean = 0

↳ std = 1

WHY IS IT NECESSARY TO CONVERT A NORMAL DISTRIBUTION INTO A STANDARD NORMAL DISTRIBUTION :-

→ Converting a normal distribution to a STANDARD NORMAL DISTRIBUTION is necessary mainly to make calculations easy and to compare things fairly across different situations.

1. A COMMON TABLE / FORMULA FOR ALL :-

• Every normal distribution has its own mean and standard deviation, so directly calculating probabilities for each one would be messy.

• By converting to $Z \sim N(0,1)$, using $Z = \frac{x - \mu}{\sigma}$, you can use a single standard normal table or norm.cdf() for all problems (marks, heights, errors, etc).

2. CLEAR INTERPRETATION OF ANY VALUE :-

• AFTER standardization, each value tells "how many standard deviations away FROM the mean" it is, so we immediately know whether it is TYPICAL, HIGH or VERY EXTREME (OUTLIER).

• This makes it easy to talk about positions like "top 10%" or "roughly average" without caring about the original units.

3. COMPARING DIFFERENT SCALES :-

• Two different exams or features (e.g., marks out of 100 vs 200, Height vs Weight) cannot be compared directly but their Z-scores can be.

• Standard normal scaling puts them on the same 0-mean, 1-sd, which is essential in data analysis and ML.

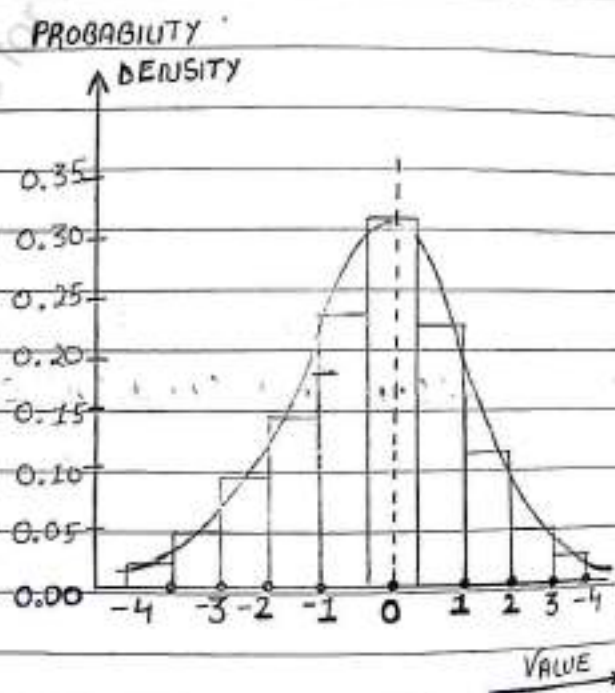
(FEATURE SCALING BEFORE TRAINING MODELS)

→ Z SCORE TELLS US how many standard deviations from the mean each value lies.

4. BUILDING MORE ADVANCED METHODS

- Many STATISTICAL TESTS (Z-TEST, Confidence Intervals) and approximations assume a standard normal distribution as base.
- In DATA SCIENCE Algorithms often work better and converge faster when inputs are standardized ($\mu=0$, $SD=1$) so this step is built into processing pipelines.

* GRAPHICAL REPRESENTATION :- STANDARD NORMAL DISTRIBUTION



- * To convert a normal distribution $X \sim N(\mu, \sigma^2)$ into a standard normal distribution $Z \sim N(0, 1)$, transform each value of X from the original distribution into a Z-score.

- * The Z-SCORE measures how many standard deviations a value x is from the mean μ .
Subtract μ from x , then divide by σ :-

$$Z = \frac{x - \mu}{\sigma}$$

→ This standardization applies to individual data points or probabilities; For example:

$P(X < x)$ becomes $P(Z < z)$ using standard normal tables

→ Example :-

Suppose $X \sim N(100, 15^2)$ and you want to standardize $x = 115$

$\therefore Z = \frac{(115 - 100)}{15} = 1$, so it lies **1 SD** above the mean.

NOTE :- A $z=1$ position on the standard normal curve $N(0,1)$ corresponds to the point where cumulative probability $P(Z < 1) \approx 0.8413$, so about **84.13%** of values are below this standardized point (versus calculating directly on the original curve).

→ WHAT DOES 84.13% OF VALUES ARE BELOW mean?

In the context of your example ($x=115$, standardized to $z=1$):

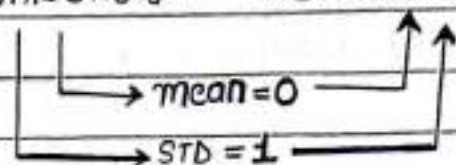
- 84.13% of data points from $X \sim N(100, 15^2) \leq 115$
- This shaded area under the curve (left tail to $z=1$) represents the proportion of the total distribution $\leq$ that point.

The remaining $\sim 15.81\%$ lies above $z=1$ (right tail)

Z-SCORE	% BELOW ($P(Z \leq z)$)	EQUIVALENT IN ORIGINAL DIST.
1	84.13%	≤ 115
0	50.00%	≤ 100
-1	15.81%	≤ 85
		CALCULATED AT DIFFERENT VALUES

STANDARDIZATION

NORMAL DISTRIBUTION $\longrightarrow$ STANDARD NORMAL DISTRIBUTION



$$z = \frac{x - \mu}{\sigma}$$

$z \in +ve$ (GREATER than μ)
 $z \in -ve$ (LESS than μ)
 $z = 0$ (Equal to μ)

NORMALIZATION: Rescales a dataset, so that each value falls between 0 and 1.

SCALING TECHNIQUE

- For example :- If I have a dataset with several numerical variable with different SCALE (1 variable $\rightarrow$ SALARY $\rightarrow$ 1000s), this creates scaling difference (2nd variable $\rightarrow$ POPULATION $\rightarrow$ lakhs), and if we perform analysis / ML Algorithms then different scale variables will influence the result differently affects the result.

$\therefore$ Therefore in such a situation we need scaling to be performed, in technical terms $\rightarrow$ NORMALIZATION

MIN-MAX SCALING / NORMALIZATION FORMULA

Give a feature x with minimum x_{min} & maximum x_{max} each value is transformed as :-

$$x_{SCALED} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

• FEATURE $\in$ VARIABLE

(HEIGHT,
AGE,
MARKS,
SALARY)

$\rightarrow$ This makes the smallest value become 0 & the largest value become 1, everything else squeezed proportionally b/w

* WHY IT IS USED ?

- Puts all FEATURES on the same scale so no single feature dominates just because it has larger numeric value.
- Useful before ML Algorithms that rely on distances or gradients (KNN, neural networks, SVM, etc)

* Example :-

Min = 3 lakhs

Max = 50 lakhs

$$x = 4 \text{ lakhs} \quad \therefore x_{\text{new}} = \frac{4 - 3}{50 - 3} = \frac{1}{47} = 0.0212 \approx 0 \leftrightarrow 1$$